Quantum migration exposes Bitcoin's deferred values choice

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Abstract

A cryptographically relevant quantum computer running Shor's algorithm could derive private keys from exposed public keys, breaking the elliptic curve digital signature algorithm that secures Bitcoin transactions. Over a third of Bitcoin's circulating supply is exposed. Bitcoin's governance structure handles routine operational and additive opt-in change well, and handles resistance to exogenous capture, censorship, and arbitrary intervention by actors hostile to Bitcoin's constitutive commitments very well. However, the configuration handles coordinated change toward an external deadline poorly. I develop a directional treatment of institutional latency that distinguishes resistance to exogenous change, coordination toward endogenous change, and coordination toward exogenous change as analytically separable directions in which a governance architecture's response capacity may differ systematically. Applied to the BIP 360 / BIP 361 case through a four-level institutional lens, this analysis locates BIP 360 at operational and implementation levels where the architecture's normal capacity is adequate, and locates BIP 361 at political and epistemic levels where the architecture is structurally resistant. The BIP 361 proposal is epistemic in nature, about what Bitcoin's commitments are for, and whether their service to that purpose is negotiable under conditions the original commitments did not anticipate. Bitcoin's architecture forces the community to answer those questions without formal deliberative tools. In this analysis, I specify the features that any subsequent formal modeling of BIP 360 / BIP 361 support would need to consider and identify candidate methodologies that could help build understanding about whether Bitcoin's post-quantum readiness was proceeding on a pathway, and at a pace, that counters threats from anticipated technological advances.

Keywords: BIP 360, BIP 361, deep uncertainty, IAD framework, institutional latency, governance, post-quantum cryptography, quantum computing, volitional pragmatism`

1. Introduction

A cryptographically relevant quantum computer (CRQC) running Shor's [1] algorithm can derive private keys from exposed public keys [2], breaking the elliptic curve digital signature algorithm that secures Bitcoin transactions [3, 4]. Over 1/3rd of Bitcoin's circulating supply has revealed a public key on-chain and is exposed to long-exposure quantum attacks once such a computer becomes available [5]. The cryptographic response is well-developed: NIST has ratified post-quantum (PQ) signature schemes [6, 7]; the Bitcoin community has drafted technical proposals for migration [5, 8]; and the engineering work to deploy quantum-resistant infrastructure is tractable [9]. The challenge is not the cryptography. I suggest, instead, that it is the institutional question that the cryptography forces: what does Bitcoin's governance require of its participants when the easy path of inaction is no longer available?

Bitcoin's governance architecture rests on permissionlessness [10] and manifests as a specific configuration of Ostrom's IAD boundary and position rules [11], with exit as the sole formally binding mechanism for implementation and political disputes. Bitcoin's governance structure handles routine operational and additive opt-in change well, and resistance to exogenous capture, censorship, and arbitrary intervention by actors hostile to Bitcoin's constitutive commitments very well. However, the protocol handles coordinated change toward an external deadline poorly.

Quantum migration cannot be purely opt-in if the threat is taken seriously; the population of UTXOs whose owners cannot or will not migrate voluntarily includes approximately 1.7 million BTC held in early Pay-to-Public-Key (P2PK) addresses[1]. Leaving those exposed could create an expansive supply-shock [12] attack surface that the institutional holders, exchanges, and protocol developers responsible for Bitcoin's continued operation cannot defensibly ignore. Bitcoin's architecture is thus being asked to do something its design did not anticipate: produce a coordinated response to an external threat on a timescale the threat dictates.

Two Bitcoin Improvement Proposals – BIP 360 [8] and BIP 361 [5] – engage Bitcoin's architecture at structurally different levels and have encountered structurally different responses. BIP 360, an opt-in PQ output type, has progressed through technical review and testnet implementation with limited controversy. BIP 361, a phased deprecation of legacy elliptic curve digital signature algorithm (ECDSA) and Schnorr signatures that ultimately freezes unmigrated coins, has generated broad public criticism alongside acknowledgment from its own authors[2] that the proposal is not currently in a position to be adopted. The differential response can be explained by what the two proposals ask the architecture to do.

In this paper, I develop a directional treatment of institutional latency that frames Bitcoin's quantum dilemma. I apply the four-level Institutional Analysis and Design (IAD) configuration to the BIP 360 /

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1 https://www.unchained.com/blog/bitcoin-address-types-compared

2 https://cointelegraph.com/news/bitcoin-devs-and-researchers-propose-freezing-quantum-vulnerable-coins-bip-361

BIP 361 case, and identify the epistemic question that BIP 361 surfaces. I treat the technical proposals as institutional artifacts whose implicit governance commitments warrant explicit examination and identify the analytical features any subsequent formal modeling would need to handle, as well as the candidate methodologies that may facilitate progress towards Bitcoin's quantum-readiness.

2. The directional structure of institutional latency

2.1. Algorithmic velocity, informational velocity, and institutional latency chain

The institutional latency concept [11] presupposes a causal chain whose elements have not always been made explicit. Algorithmic velocity refers to the rate of change in technical capability as computational capacity and intelligence scale: the pace at which technology evolves, capability frontiers move outward, and threats – and opportunities – emerge. Informational velocity refers to the rate at which information about those changes propagates through the governance-relevant ecosystem – the pace at which actors generate, process, transmit, and integrate the analytical content needed to formulate a response. Institutional latency refers to the lag between recognition and binding governance response, given the algorithmic and informational inputs.

The chain is causal in a specific sense. Algorithmic velocity sets the rate at which the substantive problem changes. Informational velocity sets the rate at which the system can perceive the change. Institutional latency is a metric that measures the gap between perception and governance response. When algorithmic velocity exceeds informational velocity, the system is operating without adequate situational awareness – actors do not know what is happening fast enough to respond appropriately. When informational velocity exceeds algorithmic velocity but institutional latency remains high, the system has the information it needs but cannot mobilize a binding response within the available window. The two failure modes are analytically distinct and have different research and governance implications.

The quantum migration case clearly illustrates the chain. Algorithmic velocity on the offensive side is high and accelerating: the CRQC timeline has compressed substantially during 2024-2026 as algorithmic improvements reduced the qubit requirements for breaking 256-bit elliptic curve cryptography by approximately an order of magnitude over the prior consensus estimate[3]. Informational velocity is high and approximately keeping pace: the threat is well-characterized in technical literature [3, 13], the cryptographic standards are nearly settled [6, 7], the BIP texts are public [5, 8], and substantial monitoring infrastructure tracks observed quantum hardware progress[4].

My argument is that institutional latency is the binding constraint. The Bitcoin governance architecture has the technical capability to deploy PQ signature schemes (algorithmic velocity is not the obstacle) and

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[3] https://security.googleblog.com/2025/05/tracking-cost-of-quantum-factori.html

[4] https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point#/

the ecosystem has the analytical content to specify what should be deployed (informational velocity is not the obstacle). What the architecture does not have is a mechanism for converting recognition into binding response on a timescale matched to the offensive curve.

The governance lag is not a function of technical or informational limits. It is a property of Bitcoin's design architecture itself – specifically, of how epistemic- and political-level tools handle questions whose answers need volitional input [14] that cannot be reached through the architecture's preferred 'code-as-law' mode of resolution. The architecture's failure mode is not that latency is uniformly high; it is that institutional latency varies systematically by the type of governance question being asked. The remainder of this section develops a directional treatment of institutional latency that helps distinguish the response capacity of governance architectures across different threat classes.

2.2. Institutional latency

The institutional latency concept [11, 15] initially focused on the basic friction between algorithmic velocity and governance response capacity as a scalar. An increasing gap between what the system needs to do and what its governance architecture can produce within the available time horizon is what defines institutional latency. This treatment is adequate for cases where governance responsiveness is approximately symmetrical across the directions of governance response. Many institutional architectures satisfy this approximate symmetry but Bitcoin does not.

The asymmetry is structural rather than incidental. The same governance properties that produce low institutional latency for one direction of response – exit dominance, no central authority, political commitments embedded in the sovereignty-first worldview [16] that relies heavily on informal social norms – produce high institutional latency for other directions. That is, the architecture is optimized for some directions of governance response and exhibits its strongest resistance in others. Aggregating across directions to a single scalar would conceal exactly the variation that determines whether the architecture can handle a given threat.

The directional structure has three implications. First, institutional latency cannot be reduced to a single scalar without losing the analytically relevant distinction between threat class. Second, the appropriate institutional latency metric depends on the direction of governance response the threat demands: a given architecture may exhibit favorable institutional latency on one direction and unfavorable institutional latency on another simultaneously, so the same architecture may be robust against one threat class while being fragile against another. Third, architectural comparison requires attention to threat-class composition rather than to aggregate latency. Two architectures with similar scalar-derived levels of institutional latency may face very different exposure profiles depending on the directions in which their latency is concentrated.

A useful three-fold distinction emerges from the Bitcoin case. The first direction is resistance to exogenous change, the architecture's capacity to refuse changes proposed by actors outside the

community, including capture attempts, regulatory imposition, and adversarial protocol modification. The second is coordination toward endogenous change, the architecture's capacity to enable changes the community itself originates and develops through its normal, largely informal processes. The third is coordination toward exogenous change, the architecture's capacity to enable changes the community must make in response to external drivers it did not create and cannot ignore. The three directions are not symmetric in the demands they place on governance.

2.3. The directional institutional latency configuration in Bitcoin

Bitcoin's exit-dominant governance structure is a textbook case of directional asymmetry. The architecture handles each of the three directions differently, and the differences map onto the threat class that has led to its current quantum dilemma.

Resistance to exogenous change is where Bitcoin's architecture performs strongest. Exit threats discipline implementation choices through fork pressure. At the implementation level, exit threats discipline soft-fork activation through user-activated soft fork (UASF) coordination, with the segwit activation history demonstrating the mechanism's effectiveness against miner attempts to capture the upgrade path through the credibility of the UASF threat rather than its deployment [17, 18]. At the political level, exit is the formally binding mechanism for resolving competing interpretations of Bitcoin's identity – peer-to-peer cash, digital gold, neutral settlement layer, etc. – and has successfully resisted multiple attempts to redefine the protocol through hierarchical authority [19]. The architecture's resistance to exogenous change is not incidental but constitutive: Bitcoin's institutional value derives substantially from its demonstrated capacity to refuse changes that would compromise the properties that its constituents value.

Coordination toward endogenous change is where the architecture performs adequately, but with characteristic delays. The community develops a BIP, refines it through public discussion and technical review, and activates it through soft-fork mechanisms when sufficient consensus exists. The Taproot upgrade illustrates the pattern: it was first proposed in early-2018, drafted as BIPs 340-342 in January 2020, and activated in November 2021, approximately four years from initial proposal to activation [20]. The timescale is long but bounded. The architecture handles this direction by allowing time-scale flexibility. The community can, in fact, afford to take years on endogenous changes because nothing external imposes a deadline.

Coordination toward exogenous change is where the architecture performs poorly and where the quantum migration question now sits. The defining feature is that the deadline is exogenous – quantum's offensive curve does not wait for the architecture's preferred deliberation timeline. The mechanisms that make the architecture robust against exogenous *unwanted* change are the same mechanisms that make it slow against exogenous *necessary* change. Exit imposes costs on actors proposing change but no costs on actors blocking change; this asymmetry is built into the mechanism and it operates without regard to whether the proposed change is hostile or defensive. A small minority can block a coordination-requiring

change by threatening to exit because the threatened split creates a costly fork the would-be majority is trying to avoid; a large majority cannot compel coordination by threatening to exit, because the would-be minority is content with the existing protocol and majority departure simply means the minority continues operating the protocol on their own terms while the majority operates whatever new protocol they prefer on a separate chain [17, 21].

This is the analytical structure that Williamson's transactional attributes framework identifies as governance form mismatch [22, 23]. The framework characterizes transactions across three attributes – asset specificity, uncertainty, and frequency – and predicts the transaction cost-minimizing governance form (market, hybrid, or hierarchy) appropriate to each combination.

The political level of the quantum defense question occupies the configuration that Williamson's framework identifies as demanding hierarchical governance: high asset specificity (Bitcoin's value is specific to its constitutive commitments), high uncertainty (about the offensive curve and about which PQ scheme will prove durable, and about Bitcoin's defense curve, which relies on uncertain technical and governance details), and low frequency (political change is rare by design). Williamson's framework predicts hierarchical governance as the transaction cost-minimizing form for this configuration. The configuration Bitcoin's quantum challenge presents, however, calls for hierarchical governance that Bitcoin is structurally incapable of supplying. The mismatch is normally tolerable because political-level demands are rare and self-imposed; quantum migration is the case that demands collective action, is recurrent (once for each successive PQ standard generation), and is imposed by exogenous technological advance in quantum computing.

3. The quantum threat requires collective action

3.1. The threat profile

A CRQC running Shor's algorithm can derive private keys from exposed public keys [2], breaking the ECDSA and Schnorr signatures that secure Bitcoin transactions. The threat is bounded by what Shor's algorithm cannot do – it does not break SHA-256 hash functions, which Grover's [24] algorithm can attack only with quadratic speedup that practical key lengths comfortably outpace [3, 25]. Bitcoin mining and the integrity of the chain itself are not the immediate concern. The immediate concern is the signature scheme that secures spending authorization for individual UTXOs.

The exposure profile is uneven. As of March 2026, more than 34% of Bitcoin's circulating supply has revealed a public key on-chain [5]. The exposure includes early P2PK addresses that include approximately 1.7 million BTC, of which approximately 1.1 million is attributed to Satoshi – along with reused addresses across all output types and Pay-to-Taproot (P2TR) outputs expose the output public key on-chain at UTXO creation (Beast et al. 2024); the internal public key is revealed only on script-path spends. Both keys become quantum-vulnerable once exposed, so the architectural protection P2TR offers

depends on whether spends use the key path or the script path. Approximately 5.6 million BTC, roughly 28% of total supply, have not moved in over a decade [5]; this dormant population is sometimes treated as lost, though recent movements of long-dormant UTXOs during 2024-2025 indicate that dormancy and loss are not equivalent, and that the share of genuinely inaccessible coins is likely smaller than the dormancy figure suggests [12].

The threat is time-bounded by the offensive curve. NIST has ratified production-grade PQ signature schemes in 2024 [6, 7], with FIPS 206 expected in draft for 2026. Algorithmic improvements during 2025 reduced the qubit requirements for breaking 256-bit ECDSA by approximately an order of magnitude over the prior consensus estimate[5]. McKinsey places CRQC capability emergence as early as 2027-2030[6]; other more conservative estimates extend the range to the mid-2030s [26]. The dispersion across credible estimates is wide and the lower bound has been moving earlier rather than later; the timeline exhibits deep uncertainty [27, 28] on multiple fronts.

A useful distinction divides the attack surface. Long-exposure attacks operate against keys exposed on-chain for arbitrary durations, giving the attacker as much time as needed to perform quantum key recovery. Short-exposure attacks operate against keys revealed only briefly during transaction confirmation, and require CRQC capability sufficient to complete the recovery within a single block confirmation interval [8]. Long-exposure attacks are the first-order threat because they require less CRQC capability and target the largest exposed value. Short-exposure attacks arrive later on the offensive curve and would require PQ signature schemes to defend against. This distinction matters for intervention design because not every intervention addresses both attack classes.

3.2. The technical proposals as institutional artifacts

BIP 360 [8], a "Pay-to-Merkle-Root" (P2MR) proposal, introduces a new opt-in output type that operates with nearly the same functionality as P2TR but removes the quantum-vulnerable key-path spend. P2MR commits to the Merkle root of a script tree without committing to an internal public key, which protects against long-exposure attacks. The proposal does not address short-exposure attacks, introduce PQ signature schemes, require any user to migrate, or constrain any user's ability to continue using existing output types. The proposal is enabling infrastructure: it adds a capability to the protocol without engaging anyone's commitments. BIP 360 entered testnet implementation in early-2026 and merged into the BIPs repository in February 2026 [for recent updates on GitHub, see 8].

BIP 361 [5], a "Post Quantum Migration and Legacy Signature Sunset" proposal, addresses the population of UTXOs that BIP 360 leaves vulnerable. The proposal presupposes that some PQ output type exists (BIP 360 or successor) and adds a phased deprecation of legacy ECDSA and Schnorr

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5 https://security.googleblog.com/2025/05/tracking-cost-of-quantum-factori.html

6 https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point#/

signatures. Phase A, activating approximately three years after BIP 361 itself, prevents new sends to quantum-vulnerable addresses. Phase B, activating two years after Phase A, invalidates legacy signatures at the consensus layer, rendering coins in unmigrated addresses permanently unspendable. A possible Phase C would allow recovery of frozen funds through zero-knowledge proofs of seed phrase ownership. The proposal is enforcement infrastructure. It does not add a capability but removes one, on a published timeline, from anyone who fails to migrate.

Table 1. BIP 360 and BIP 361 compared across technical and institutional dimensions

Dimension	**BIP 360 (P2MR)**	**BIP 361 (Sunset)**
Proposal type	Enabling infrastructure	Enforcement infrastructure
Coercion	Opt-in	Mandatory
Adds or removes capability	Adds	Removes
IAD levels engaged	Operational, implementation	Political (constitutional), epistemic (meta-constitutional)
Threat coverage	Long-exposure attacks only	Full migration of vulnerable UTXOs
Target UTXO scope	Future and migrating UTXOs	All vulnerable UTXOs by Phase B flag-day
Lead authors	Beast, Heilman, & Foxen Duke (2024)	Lopp et al. (2026)
Status (April 2026)	Merged in BIPs repository; testnet implementation	Draft; informational status; pre-adoption
Community response	Limited controversy	Substantial public criticism
Architecture-fit	Within normal capacity	Forces engagement at most resistant levels

The two proposals are not technically substitutable. BIP 360 alone leaves the long-exposure problem unaddressed for the substantial UTXO population whose owners cannot or will not migrate voluntarily. BIP 361 alone is technically inoperative because it depends on a PQ output type for migration to be possible. The proposals are structurally different in their institutional demands. BIP 360 operates within the architecture's normal capacity for additive opt-in change, while BIP 361 forces engagement with the architecture's most resistant levels. The institutional question implicit in the joint proposal is at what point the marginal cost of voluntary-only migration (BIP 360 plus operational and implementation infrastructure) exceeds the marginal cost of coercive migration (BIP 360 plus BIP 361).

The community's response since February 2026 illustrates the directional asymmetry the framework predicts. BIP 360 has progressed through technical review and testnet implementation with limited controversy. BIP 361 has generated substantial public criticism[7]. The proposal has been described in community discourse as "authoritarian," "laughable," and "confiscatory," alongside acknowledgment

[7] https://cointelegraph.com/news/bitcoin-devs-and-researchers-propose-freezing-quantum-vulnerable-coins-bip-361

from the proposal's own authors that BIP 361 "isn't a spec, nor is it proposed for activation. It's a rough idea for a contingency plan that needs more R&D"[8]. The differential response can be explained by the differential governance level the two proposals engage.

3.3. Bitcoin's situation in cross-domain perspective

Transport Layer Security (TLS) migration to PQ cipher suites is being coordinated through Internet Engineering Task Force (IETF) working groups with explicit standards processes and version negotiation mechanisms designed for protocol evolution[9]. Banking infrastructure migration is coordinated through regulatory agencies and SWIFT-level standards bodies whose authority extends across the institutional participants [29]. Government public key infrastructure (PKI) migration is coordinated through NIST and the Cybersecurity and Infrastructure Security Agency (CISA) at the USA federal level [30], with explicit deadlines: the Commercial National Security Algorithm Suite 2.0 [31] mandates PQ migration by 2030 with full deployment by 2033, and NIST IR 8547 plans to prohibit elliptic curve cryptography (ECC, the family that includes both ECDSA and Schnorr) after 2035 [32]. Each of these architectures supplies hierarchical coordination at the political level. The architectures are not perfectly suited to exogenous coordination because banking PQ migration is widely viewed as moving slowly and unevenly across jurisdictions [33], but they have institutional venues in which coordination can be conducted and binding decisions can be reached.

Bitcoin's situation is distinctive not because the threat is different but because the architecture lacks the coordination mechanisms the other domains take for granted. The architecture's design intent traces to the cypherpunk movement of the late-1980s and 1990s [34, 35], whose participants were deeply skeptical of state authority, monetary policy, and centralized intermediation, and who therefore preferred individual cryptographic sovereignty and mutual cooperation over hierarchical coordination as the basis for collective action. The sovereignty-first orientation persists in part of the contemporary Bitcoin community [36] and still shapes how the architecture handles questions whose answers depend on coordinated change. The cypherpunk founders probably did not anticipate, and the architecture they built was not designed to anticipate, the conditions under which exogenous pressure from technological advance would impose coordination demands the architecture's voluntary mechanisms could not satisfy. The current quantum challenge is testing a property Bitcoin's architecture was deliberately designed not to have [10, 19].

The cross-domain contrast extends to the standing of standards bodies. TLS, banking, and government PKI migrations treat NIST cryptographic standards as authoritative, while the Bitcoin community's relationship to NIST is more complicated. Bitcoin's design choices already encode precautionary stance toward NIST primitives: secp256k1 was chosen over NIST P-256 at protocol inception, and the protocol

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8 https://x.com/lopp/status/2044406134178795748

9 https://www.ietf.org/archive/id/draft-ietf-tls-hybrid-design-13.html

uses double-hashing for proof-of-work and parts of the transaction commitment structure. The skepticism is grounded in documented compromises in earlier NIST outputs, notably the Dual_EC_DRBG generator that NIST formally withdrew in 2014 after the disclosure of NSA involvement in its parameter selection [37]. The cypherpunk-aligned portion of the community treats NIST FIPS 206 finalization as a provisional input warranting independent validation rather than an authoritative standard, adding a values dimension to the migration choice the hierarchical coordination domains do not face.

4. Institutional configuration across IAD levels

4.1. Four-level analysis

The directional institutional latency framework locates Bitcoin's quantum dilemma at the intersection of high algorithmic velocity on quantum computing's offensive curve, high informational velocity in the surrounding ecosystem, and high institutional latency on coordination toward exogenous change. I work out where in the architecture that latency concentrates by working through the four levels of the IAD framework – epistemic, political, implementation, and operational – using my own more accessible terminology in place of the original meta-constitutional, constitutional, and collective-action labels [38, 39].

The two principal proposals – BIP 360 [8] and BIP 361 [5] – engage the architecture at structurally different levels (Table 2). BIP 360 operates at the operational and implementation levels and does not engage political or epistemic commitments. BIP 361 forces engagement at the epistemic level and, thus, at all IAD levels below, including the political level where narratives, resource allocation, and rule-setting privileges are contested.

Table 2. Quantum migration interventions located at the IAD levels they engage. "Primary" indicates the level at which the intervention's principal governance demand is concentrated; "Secondary" indicates a level at which the intervention also engages but where the demand is derivative or instrumental; " – " indicates the intervention does not meaningfully engage the level).

Intervention	Epistemic	Political	Implementation	Operational
User education campaigns	–	–	–	Primary
Wallet PQ output support	–	–	–	Primary
Exchange and custodian PQ migration	–	–	–	Primary
Hardware wallet firmware updates	–	–	–	Primary
BIP 360 activation (P2MR output type)	–	–	Primary	Secondary
Subsequent PQ signature scheme BIP	–	–	Primary	Secondary
BIP 361 Phase A (block sends to vulnerable addresses)	–	Secondary	Primary	Secondary
BIP 361 Phase B (invalidate legacy signatures)	Primary	Primary	Secondary	Secondary
BIP 361 Phase C (ZK proof recovery)	Primary	Primary	Secondary	Secondary

Hourglass-style throttling for P2PK	Secondary	Secondary	Primary	Secondary

4.1.1. Epistemic level

The epistemic level holds the value-based commitments that shape what Bitcoin's protocol exists to serve. These are not rules in the operational or implementation sense; they are the commitments that the rules at lower levels are designed to express. For Bitcoin, the epistemic-level commitments include immutability, bearer instrument property rights, permissionlessness, peer-to-peer money, and the broader purpose of being a credible long-term store of value outside the control of any sovereign authority. When the lower levels of the IAD framework specify what is permitted, required, or forbidden, they do so by reference to these commitments – high-level mandates – rather than independently of them.

The volitional pragmatist question [40] operates at this level. What there is most reason to want, given evolving conditions that did not exist when the original commitments were made, cannot be reduced to either reflexive defense of the original commitments or reflexive acceptance that external pressure overrides them. The pragmatist answer requires deliberative engagement with the genuine novelty of the situation [14, 40, 41] , which means working out, through inquiry, what the community now has most reason to want of an institution that was not originally designed to handle the situation it now faces.

Bitcoin's architecture provides no formal venue for this kind of deliberation. The closest analog is the protocol's informal norm layer [11], where competing identity narratives – peer-to-peer cash, digital gold, neutral settlement layer – accumulate community investment and persist or erode through diffuse mechanisms. The norm layer aggregates expressed preferences and rewards persistent advocacy; it does not produce the structured, deliberative engagement with novel conditions that pragmatist inquiry calls for.

BIP 361 forces engagement at this level. The proposal's rationale invokes Bitcoin's purpose as a credible long-term store of value as the basis for restricting individual sovereignty over UTXOs whose owners may not have consented to migration. The argument is epistemic in the strict sense: it claims that Bitcoin's commitments to immutability and individual sovereignty are contingent on serving the broader purpose of being a credible long-term store of value, and that when the two come into conflict, the broader purpose takes precedence. The principled libertarian objection – that the proposal is "confiscatory" and violates Bitcoin's foundational commitments – is also epistemic in nature, though it is most often framed as a political-level claim about what the protocol's rules permit. Both positions are working at the epistemic level over what the constitutive commitments are for; neither has the institutional venue in which the question can be resolved.

4.1.2. Political level

The political level is where epistemic-level commitments are operationalized through contestation over rule-making authority, resource allocation, and norm-shaping narratives. What the epistemic level holds

as a value commitment, the political level translates into mandates about how, and which, position, boundary, authority, aggregation, scope, information, and payoff rules [38] are implemented. The contestation at this level concerns how the commitments are to be expressed in concrete protocol rules, how resources are allocated to support particular operationalizations, and which narratives shape the interpretation of the lower levels.

The directional flow runs downward. Epistemic commitments shape political contestation; political contestation shapes implementation rules; and implementation rules specify incentives that affect operational behavior. The reverse direction also operates but signals are comparatively weak [42, 43]: changes in operational behavior generate pressure on political contestation, and political contestation on epistemic commitments. The upward flow is, however, slower and works through mechanisms such as shifting norms and narratives rather than through binding mechanisms.

For the quantum case, the political level handles the contested operationalization of whatever the community works out at the epistemic level about what immutability and property rights commitments require. If the epistemic-level question of what the commitments are for were settled in favor of strict immutability, the political contestation would concern how to operationalize that strict reading – which signature schemes can be added, what voluntary migration mechanisms are acceptable, and how to handle the population of UTXOs whose owners cannot migrate. If the epistemic-level question were settled in favor of credible long-term store-of-value supremacy, the political contestation would concern how to operationalize the resulting protections – what migration timelines are reasonable, what enforcement mechanisms are proportionate, and what recovery options are required to prevent inadvertent harm. The political level cannot resolve the epistemic question; it can only operationalize whatever resolution the epistemic level produces.

Bitcoin's architecture handles the political level with characteristic difficulty when epistemic-level commitments are themselves under contestation, as they are in the BIP 361 case. There is no central authority that can issue an authoritative interpretation of the protocol's constitutive commitments, and the political-level contestation over operationalization therefore proceeds without the epistemic ground being stable. Multiple identity narratives generate different answers to the political-level questions because they rest on different epistemic-level commitments, and the architecture provides no formal mechanism for adjudicating among them. The constitutional record – the original Nakamoto [10] white paper, the accumulated body of BIPs and their associated discussions[10], the implemented protocol code[11], and the diffuse community discourse through which interpretations are formed and contested, often on social media – supplies no binding answer because the question concerns a situation the original commitments did not anticipate.

[10] https://github.com/bitcoin/bips
[11] https://coin.dance/nodes

Williamson's transactional configuration [22, 23] operates at the political level with full force. Bitcoin's value is specific to its constitutive commitments and any modification of those commitments potentially compromises the value the network exists to preserve, implying maximum asset specificity (Table 5). The quantum offensive curve is deeply uncertain, the optimal defensive response is uncertain, and the long-term consequences of any political-level decision are uncertain, so uncertainty is also at the maximum. The architecture treats political questions as exceptional and provides no formal mechanism for handling them, so frequency is, conversely, at the minimum. Williamson's framework predicts hierarchical governance for the configuration of high specificity, high uncertainty, and low frequency. Bitcoin's architecture is structurally incapable of supplying it because the architecture explicitly rejects the hierarchical authority that could supply binding deliberation. That leaves the political contestation to be conducted using *ad hoc* engagement, informal discourse, and social norms, and – moreover – to be conducted while the epistemic ground beneath it is itself unsettled.

Table 5. Williamson's transactional attributes mapped across the four IAD levels, with rationales for each level's profile and the architectural fit it produces in Bitcoin.

IAD level	Asset specificity	Uncertainty	Frequency	Rationale for the level's profile
Epistemic	High	High	Very low	Constitutive commitments are highly specific to Bitcoin's institutional value, must hold across unanticipated future conditions, and are rarely revised; to minimize the transaction costs of governance, the configuration demands hierarchical governance structure the architecture cannot supply.
Political	High	High	Low	Operationalization of constitutive commitments inherits their high specificity, must accommodate evolving operationalization conditions, and proceeds rarely and in response to exceptional pressure; the configuration also demands hierarchical governance and produces the architectural mismatch most acutely.
Implementation	Moderate	Moderate	Moderate	Specific protocol rules carry moderate specificity (revisable through soft forks), must accommodate technical evolution, and update periodically through BIP processes; the configuration is within the architecture's normal capacity, with characteristic delays.
Operational	Low	Low	High	Operational behavior is largely substitutable across implementations, governed by current rules rather than future contingencies, and occurs continuously; the configuration is handled cleanly through normal market mechanisms and exit-disciplined provider competition.

The community's response to BIP 361 in the months following its February 2026 assignment illustrates the dynamics. Substantive engagement has occurred – the proposal's authors include experienced BIP contributors, the technical community has reviewed the proposal in detail, and the public response has been vigorous and largely critical. What has not occurred, and cannot occur within the architecture, is binding adjudication. The proposal cannot be approved or rejected by any authority; it can only accumulate or fail to accumulate the diffuse community support needed for implementation-level activation. The political contestation does not get resolved by the response; it gets deferred while the epistemic question beneath it remains unsettled and the operational and implementation work on BIP 360 proceeds in parallel.

4.1.3. Implementation level

The implementation level is where political-level mandates get translated into concrete protocol upgrades, such as soft forks, BIP activations, and signature scheme deployments that operationalize whatever goals the political level has settled on. The level handles its work imperfectly, with characteristic delays. The Taproot history establishes the benchmark [20]: roughly four years from initial BIP draft to activation, with the bulk of that time spent in technical refinement, community discussion, and activation-mechanism debate. The implementation level can produce binding response but the timescale is long and the process does not accelerate well under external pressure.

Two features of the implementation level matter for the quantum case. First, the sequential structure: quantum migration requires not a single soft fork but a sequence: a PQ output type, a PQ signature scheme, possibly hybrid schemes [44], and possibly enforcement mechanisms [5, 8]. Each soft fork is a separate implementation problem with its own activation timeline, and the aggregate timeline reflects how the political-level contestation cascades into implementation-level rule-setting. Second, the parameter selection problem: multiple PQ signature schemes are available, with NIST FIPS 206 still in draft as of early-2026, and the implementation mechanism must select among them under conditions where the optimal choice is not yet settled in the broader cryptographic community.

BIP 360 was merged into the official Bitcoin Improvement Proposal repository on 11 February 2026, with a working testnet implementation activated on the separate Bitcoin Quantum network on 19 March 2026. Both milestones reflect movement at the operational and implementation levels of the architecture; neither requires the political or epistemic engagement that BIP 361 forces

4.1.4. Operational level

The operational level is where the implementation-level rules govern actual behavior – individual users migrating to PQ output types, wallet software handling the new addresses, exchanges and custodians supporting PQ migration in their infrastructure, hardware wallet firmware accepting the new signature

schemes, etc. The architecture handles this level cleanly because operational change happens through normal market mechanisms rather than through governance processes.

Cost structure at the operational level is engineering and coordination cost concentrated on a relatively small number of infrastructure providers, including wallet developers, major exchanges, custodian platforms, and hardware manufacturers. The number of organizations whose actions matter is in the dozens rather than the millions, which keeps coordination tractable. The directional institutional latency profile here is favorable across all three response categories: no external actor is forcing operational-level change; coordination toward endogenous change operates through normal market timescales; coordination toward exogenous change is also tractable because providers can move independently and partial coverage is functional even if not complete. The operational level is where the framework's analytical pressure is least, and most of the difficulty rises to the surface at higher levels.

4.2. Implicit assumptions in the BIP 361 case

The BIP 361 rationale [5] advances a sequence of analytical claims, framed as self-evident, that warrant explicit examination at the political and epistemic levels. Each claim corresponds to a question on which reasonable analysts can differ, and each is doing analytical work in the proposal that becomes visible only when stated as a substantive position rather than a background assumption. The six positions below are not exhaustive but those whose explicit statement most directly illuminates what the proposal asks of Bitcoin's architecture.

4.2.1. The "private incentive to upgrade" framing

The proposal describes Phase B's freezing mechanism as a "refined private incentive" and treats frozen unmigrated coins as economically equivalent to the Satoshi-described "donation to everyone" [12]. This conflates two cases that institutional economics distinguishes. The first is coins frozen because owners are dead, lost their keys, or are unaware – the modal case for up to 5.6-M BTC that have not moved in over a decade. The second is coins frozen because owners chose not to migrate despite capability – a small share of total frozen value but a different normative case. The first is a transfer from heirs and estates to remaining holders; the second is a Pareto-relevant signal of preference. Treating them identically aggregates over distributional consequences institutional economics typically holds open.

4.2.2. The "redistribution dilemma" trichotomy

The proposal presents the alternatives for handling lost-coin quantum vulnerability as: allow anyone to steal; allow throttled theft; or allow no one to steal. The trichotomy presupposes that prevention of theft is the dominant normative objective. An alternative framing prioritizes minimization of total welfare loss

[12] https://satoshi.nakamotoinstitute.org/posts/bitcointalk/131/

inclusive of users frozen out by mistake, which justifies partial recovery mechanisms – the Phase C zero-knowledge-proof option, the hourglass-style throttling proposal – even at the cost of permitting some quantum theft. The trichotomy is analytically restrictive in a way the proposal does not acknowledge.

4.2.3. The five-year flag-day

Phase B activates two years after Phase A, which itself activates approximately three years after BIP 360, giving a five-year lag from BIP 360 activation to legacy signature invalidation. The proposal asserts this timing as “the best balance” between giving account owners migration time and maintaining ecosystem integrity. This is the parameter on which formal decision analysis would have most to contribute. At what assumed CRQC arrival distribution does five years dominate three years or seven years? How robust is that dominance to specification of the loss function? The proposal’s five-year choice is asserted, not derived.

4.2.4. The fiduciary duty claim

Stakeholder incentives (Table 2 in the BIP 361 proposal) suggest that institutional holders face fiduciary duty to migrate, framing inaction as a duty violation. The claim is presented as legal-ethical fact but is contingent on the institution's assessment of quantum probability. An institutional holder that assesses the offensive curve as slow can defensibly meet fiduciary obligations through monitoring rather than migration; an institutional holder that assesses the curve as fast can defensibly migrate immediately. The fiduciary duty is real but its operational implications depend on the probability assessment, which the proposal treats as settled. The structural pattern parallels the long-standing adaptation-mitigation divide in climate policy [45-47], where mitigation advocates argue for maximum upfront commitment to avoid catastrophic outcomes and adaptation advocates argue that resources committed prematurely under high uncertainty may be diverted from higher-return alternatives. BIP 361 mirrors the mitigation-advocate divide implicitly, treating maximum upfront migration as the duty-satisfying response regardless of the probability assessment

4.2.5. The unilateral stakeholder framing

The stakeholder incentives table also lists incentives to upgrade but not the costs of premature upgrade. A symmetrical analysis would acknowledge: engineering costs at exchanges and wallet providers; fee and throughput costs from larger PQ signatures; coordination risk if the selected scheme is later deprecated in favor of a successor (e.g., the forthcoming FIPS 206 finalization may favor a scheme not currently implemented); and option value lost by committing to a particular migration path before the technical landscape settles. The unilateral framing parallels patterns documented in regional water infrastructure planning under deep uncertainty [48, 49], where centralized supply-side planning that does not account for distributed demand-side adaptation produces capital over-allocation; BIP 361's mandatory

enforcement risks the analogous over-allocation if the bottom-up migration that BIP 360's voluntary mechanisms would generate is underestimated.

The option-value cost is the most consequential of these because it is the one most directly addressed by the BIP 360 / BIP 361 timing choice itself. Committing to a specific PQ scheme before NIST FIPS 206 finalization and IETF/TLS settlement forecloses the option to migrate to whatever scheme the eventual ecosystem convergence selects, at engineering and coordination cost the proposal does not price

4.2.6. The status quo specification

The proposal evaluates BIP 361 against an implicit *status quo* of inaction. At least three plausible candidates for the appropriate *status quo* exist: current state with no PQ infrastructure; BIP 360 activated as a floor with no enforcement; and BIP 360 plus voluntary best-practice migration without enforcement. The "optional now" position articulated by Bitcoin developer Adam Back[13] occupies the second or third of these. Each generates a different counterfactual against which BIP 361's marginal contribution is evaluated. A differential evaluation across these candidate baselines is what formal CBA would produce, and the result depends materially on which counterfactual is taken as the baseline.

These six implicit positions do not invalidate the BIP 361 proposal. They identify where the proposal's economic claims rest on substantive, but unstated, normative and empirical commitments. A formal institutional economics treatment would identify each commitment, examine its sensitivity to assumption, and characterize the conditions under which the proposal's policy conclusion follows.

4.3. IAD interpretation

The four-level analysis predicts that Bitcoin's architecture handles the operational and implementation levels with characteristic adequacy and the political and epistemic levels with characteristic difficulty. The BIP 360 and BIP 361 proposals split along this divide: BIP 360 operates entirely at the operational and implementation levels and is moving through the architecture at the architecture's normal pace, while BIP 361 forces engagement at the political and epistemic levels, and is encountering the architecture's formidable resistance at those levels. The institutional dilemma is not that the architecture has failed; it is that the architecture is doing exactly what its directional institutional latency profile predicts, and what the architecture does at the higher levels is not adequate to the time horizon for the looming threat.

The directional profile at the epistemic level is the most unfavorable in the architecture. Resistance to exogenous change is institutionally expressed as the refusal to allow external conditions to revise constitutive commitments – a position with strong principled support and substantial historical basis in Bitcoin's governance development process [11, 19]. Coordination toward endogenous change at the

[13] https://www.coindesk.com/tech/2026/04/16/bitcoin-s-quantum-debate-splits-as-adam-back-pushes-optional-upgrades-over-forced-freeze

epistemic level is rare and slow because the architecture treats such coordination as suspect by design. Coordination toward exogenous change at the epistemic level is what BIP 361 ultimately requires and what the architecture is least equipped to supply. The governance framework cannot specify how change should be supplied; specifying that would require either a hierarchical authority Bitcoin does not have or non-coercive and timely deliberative mechanisms that have not been developed. The framework's diagnostic contribution is to clarify why the question is hard, which is the necessary first step toward whatever pragmatic process the community ultimately conducts to address it.

That is, this paper locates the dispute correctly without resolving it. BIP 361's authors and its critics are not differing on technical cryptography, on the existence of the threat, or on the desirability of network security; all of these are mainly settled. They are differing on what Bitcoin's commitments are for and whether the answer to that question can be revised because of emergent external pressure. Identifying the actual subject of the dispute at this epistemic level is what the institutional latency framing offers. The quantum challenge can be conducted productively only if the parties recognize that they are arguing about the epistemic question, because the political and implementation mechanisms cannot supply an answer until the epistemic question is, at least provisionally, settled.

5. Research needs and opportunities

5.1. Hypotheses on Bitcoin's response to the quantum challenge

The framework generates testable predictions about the Bitcoin community's response to the quantum challenge over the coming years. The hypotheses are framed as conditional statements with specified mechanisms and empirical indicators rather than as predictions in the strong sense. Each hypothesis identifies what the framework would lead one to expect, conditional on the framework's analytical structure being correct, and specifies how the prediction would be empirically adjudicated.

Table 3. Hypotheses on Bitcoin's response to the quantum challenge

Hypothesis	Mechanism	Empirical indicator
H1. BIP 360 will progress through the architecture substantially faster than BIP 361, controlling for technical complexity.	Directional institutional latency: BIP 360 operates at operational and implementation levels where the architecture handles change adequately; BIP 361 forces engagement at political and epistemic levels where the architecture is structurally resistant.	Time from BIP draft to activation for BIP 360 versus time to non-rejection or activation for BIP 361.
H2. Bitcoin community sentiment on coercive migration mechanisms will shift non-linearly rather than linearly in response to substantive offensive-curve updates.	Epistemic engagement: incremental updates to CRQC capability estimates produce small political-level adjustments; threshold-crossing demonstrations produce discrete epistemic reassessments.	Comparison of community sentiment shifts to incremental CRQC capability updates versus discrete threshold-crossing demonstrations or near-demonstrations.

H3. The institutional holder population (ETFs, treasuries, sovereign holders) will diverge from the self-custody population in BIP 361 sentiment, with institutional holders becoming supportive earlier than the offensive-curve probability alone would predict.	Constituency formation: stability-first [16] institutional holders face fiduciary and reputational pressures that compress their decision horizon; sovereignty-first self-custody holders face no equivalent compression.	Differential sentiment in institutional research notes, custody-platform statements, and ETF prospectus disclosures versus self-custody community discourse and developer-forum positions.
H4. Any actual demonstration of quantum capability against ECDSA at sub-Bitcoin key lengths will produce community responses qualitatively different from incremental CRQC capability claims of equivalent technical magnitude.	Threshold-crossing salience: epistemic questions deferred under conditions of capability uncertainty become difficult to defer under conditions of demonstrated capability, regardless of the gap between demonstrated and threatening capability levels.	Comparison of community discourse, BIP-process activity, and institutional research output before and after any verified ECDSA break at any key length.
H5. Within the institutional / self-custody divergence predicted in H3, BIP 361 scheme-commitment timing preferences will further differentiate by trust in NIST cryptographic standardization, with sovereignty-first holders favoring delayed commitment pending independent validation.	Differential trust in standards bodies: stability-first holders accept NIST standards as operationally necessary; sovereignty-first holders inherit skepticism rooted in documented NIST compromises [37] that Bitcoin's design choices already encode.	Comparison of scheme-commitment timing preferences in custody-platform statements and ETF disclosures versus self-custody community discourse, controlling for the H3 institutional/self-custody split.

The five hypotheses concern, respectively: the differential pace of BIP 360 and BIP 361 progression through the architecture (the directional institutional latency prediction); the non-linearity of community sentiment shifts in response to offensive-curve updates (the epistemic engagement prediction); the divergence between institutional and self-custody populations in BIP 361 sentiment (the constituency formation prediction); the architecture's response to any actual demonstration of quantum capability against ECDSA at sub-Bitcoin key lengths (the threshold-crossing prediction); and the further differentiation of scheme-commitment timing preferences by trust in NIST standardization within the institutional / self-custody divergence (the differential standards-trust prediction). Each hypothesis is specified at a level of generality that permits empirical adjudication within a 2- to 5-year horizon, conditional on observable indicators that can be tracked through public discourse, BIP-process records, and new research outcomes.

5.2. Research agenda

Translating the Table 3 hypotheses into analyses that provide policy-salient evidence requires moving from the descriptive treatment above to formal modeling that captures cost structures, option values, and the deep uncertainty in CRQC arrival timing. The candidate methodologies fall into three clusters: traditional analytical options inherited from environmental and financial economics; decision-making under deep uncertainty (DMDU) approaches developed for deeply uncertain policy challenges (e.g.,

terrorism threats, national security, environmental change and management); and hybrids that integrate elements of both (Table 4).

Table 4. Candidate analytical methodologies for formal analysis of the BIP 360 / BIP 361 choice

Methodology	Cluster	Analytical capability	Limitation for the Bitcoin case
Cost-benefit analysis (CBA)	Traditional	Transparent ranking by net benefit under specified assumptions	Fails on irreducible uncertainty in cost components
Real options analysis	Traditional	Values the option to defer commitment under information arrival	Assumes diffusive uncertainty and instantaneous option exercise
Marginal abatement cost curves (MACCs)	Traditional	Cost-effectiveness ranking with graphical apparatus for decision-makers	Static apparatus; rank-ordering distribution-sensitive
Robust decision making (RDM)	DMDU	Stress-tests strategies across thousands of plausible futures; identifies robust strategies that minimize regret	Does not naturally handle dynamic sequencing
Dynamic adaptive policy pathways (DAPP)	DMDU	Pathway maps with tipping points; handles sequential intervention dynamics	Depends on identifiable tipping-point indicators
Info-gap decision theory	DMDU	Robustness of decisions to deviations from a central estimate	Single-parameter focus; multi-stakeholder welfare aggregation harder
Combined DAPP-MACC	Combined	Offense and defense MACCs integrated through dynamic pathway analysis	Methodologically composite; novel modeling approach with no current implementation

5.3. Traditional analytical options

5.3.1. **CBA**

Standard cost-benefit analysis (CBA) [50] with point estimates ranks candidate interventions by net benefit under a specified set of assumptions about costs, benefits, discount rates, and the CRQC arrival distribution. Applied to the Bitcoin case, CBA would specify the cost structure of each intervention – user education campaigns, wallet PQ support, exchange and custodian migration tooling, hardware wallet firmware updates, BIP 360 activation, BIP 361 Phase A and B activation, hourglass-style throttling for P2PK addresses, and Phase C zero-knowledge recovery mechanisms – and produce expected-value comparisons. The methodology is transparent and widely understood but it fails on the irreducible uncertainty in several cost components. The option cost depends on the eventual NIST and IETF settlement on PQ signature schemes; the coordination cost depends on the architecture's response to its own epistemic question; and the user friction cost depends on adoption dynamics that no current data

calibrates well. Point estimates cannot honestly summarize this uncertainty and methodologies that require them produce false precision [27].

5.3.2. **Real options**

Real options analysis from financial economics [51] values the option to defer commitment under conditions where new information arrives stochastically and where premature commitment forecloses options that later information would have made attractive. Applied to the Bitcoin case, real options analysis would, for example, value the decision to activate BIP 361 today rather than two years from now, given that NIST FIPS 206 finalization, empirical evidence on PQ signature performance, observed quantum hardware progress, and observed migration adoption rates conditional on BIP 360 alone may all arrive within the waiting period.

The standard apparatus applies imperfectly. The underlying uncertainty is not the diffusive Brownian motion that financial real options models assume but it is structured uncertainty about the timing of a discrete capability emergence. The countervailing consideration is that coordination capacity itself depreciates during the waiting period. Bitcoin's architecture can takes years to mobilize a political-level response [e.g., 20] and the period of mobilization must precede the period of action by an architectural lag that compounds the option value calculation. The standard real options analysis treats the option exercise as instantaneous [52] but in the Bitcoin case the option exercise itself takes years and must be initiated well before the option is to be exercised. The effective deadline for beginning epistemic and political deliberation is therefore earlier than the deadline for completing the migration, by a margin equal to the architecture's institutional latency gap.

This is a genuine extension of the real options framework. The Bitcoin application asks when to begin a deliberative process whose completion is required for the commitment to be possible at all, pricing both the option value of waiting for technical information and the option-destruction cost of beginning deliberation too late for it to complete in time. The two effects pull in opposite directions.

5.3.3. **MACC analysis**

Marginal abatement cost curves (MACCs), introduced for climate carbon mitigation analysis [53, 54], rank interventions by cost-effectiveness and produce a graphical apparatus that decision-makers can read directly. Applied to the Bitcoin case, MACCs would decompose the cost side of each intervention into engineering, coordination, user friction, fee/throughput, and option components, and the benefit side into the percent of UTXO value protected, the attack surface reduced (long- versus short-exposure), and the coordination uncertainty resolved. Industry technical reports analogous to the methane MACCs [e.g., 55] provide the precedent for fine-grained intervention-by-intervention cost decomposition that the Bitcoin case would require.

The MACC apparatus does not handle the deep uncertainty in CRQC arrival distributions natively. The rank-ordering of interventions may shift substantially across plausible CRQC arrival distributions. Under late-arrival distributions, interventions with high option costs lose attractiveness. The value of waiting for further information remains an important factor in the cost-benefit calculus and the BIP 360-only path dominates because it preserves flexibility for the eventual signature scheme selection. Conversely, under early-arrival distributions the option value of waiting evaporates; the BIP 360 plus BIP 361 path would dominate because the cost of being unprepared exceeds the cost of premature commitment. The CRQC arrival distribution is exactly the parameter on which the most informed observers disagree most sharply and the rank-ordering of interventions inherits that disagreement. A static MACC analysis must therefore be supplemented with explicit treatment of the distribution itself, which moves the analysis into the DMDU world.

5.4. DMDU modeling options

The traditional analytical options each fail on at least one feature of the Bitcoin case: CBA on the irreducible cost-component uncertainty; real options on the coordination-capacity extension; and MACCs on the distribution-sensitivity of the rank-ordering. The DMDU literature has developed methodologies designed for problems where model structure is uncertain, the relevant probability distributions cannot be reliably specified, and where the analysis must accommodate genuine uncertainty rather than assume it away.

5.4.1. **RDM**

Robust decision making (RDM), developed at RAND [28], inverts standard decision-analytic logic by stress-testing candidate strategies across many (e.g., hundreds of thousands) of plausible futures and identifying which perform acceptably, minimizing regret across the widest range of scenarios. The methodology integrates ensemble model runs, scenario discovery techniques, and segmentation analyses to identify the parameter regions in which each candidate strategy fails or succeeds [48, 56]. Applied to the Bitcoin case, RDM would specify a small set of migration strategies – BIP 360 alone; BIP 360 plus BIP 361 on the proposed timeline; BIP 360 plus BIP 361 with Phase C; partial-recovery alternatives – and characterize the conditions under which each strategy fails catastrophically or performs robustly across the joint parameter space of CRQC arrival distribution, signature scheme durability, migration adoption rate, and architectural coordination latency. The methodology handles the irreducible uncertainty in the marginal cost decomposition by treating uncertain parameters as exploratory dimensions rather than as quantities requiring point estimates. It does not, however, naturally handle the dynamic sequencing the BIP 360 / BIP 361 case presents.

5.4.2. **DAPP analysis**

Dynamic adaptive policy pathways (DAPP), developed for water management problems facing deep uncertainty about climate-driven capability thresholds [49, 57], is built around adaptation tipping points and the conditional sequencing of interventions as information arrives. Applied to the Bitcoin case, DAPP would generate a pathway map structured around BIP 360 and BIP 361 activation. That would allow migration with periodic reassessment, and enforcement or partial-recovery alternatives as competing pathways with explicit tipping points where one strategy expires and another becomes viable.

Candidate tipping-point indicators for the Bitcoin case are reasonably identifiable: NIST FIPS 206 finalization and the specific PQ signature scheme selected; verified offensive-curve thresholds and demonstrated quantum factorization at increasing key lengths; institutional migration adoption rate; and the IETF and TLS deployment trajectories that signal whether the broader cryptographic ecosystem has converged on a specific PQ scheme. The methodology handles the sequencing problem and, through its tipping-point logic, handles the structured uncertainty about discrete capability emergence directly. It depends on the availability of identifiable tipping-point indicators with sufficient precision to support timely strategy shifts, which the Bitcoin case may or may not supply at the precision DAPP applications typically assume – though the candidate indicators are themselves the object of monitoring infrastructure already in place [26][14].

5.4.3. **Info-gap theory**

Info-gap decision theory [58] treats deep uncertainty as a horizon of unknowns around a best estimate and asks how robust a candidate decision is to deviations from the estimate, inverting the standard probabilistic framing that requires reliable distributions. Applied to the Bitcoin case, info-gap would characterize how far the CRQC arrival distribution can deviate from any specified central estimate before each candidate strategy becomes inferior to its alternatives, directly addressing the option value of waiting under deep uncertainty about the offensive curve.

The methodology is particularly well-suited to parameter-specific robustness questions – for instance, how many years the central CRQC arrival estimate can be wrong before BIP 361's five-year flag-day specification dominates three-year or seven-year alternatives – but is less suited to multi-stage sequencing or multi-stakeholder welfare aggregation. Info-gap is best understood as a complement to the broader analytical apparatus rather than as a stand-alone framework for the BIP 360 / BIP 361 choice

[14] Also see https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point#/

5.4.4. **Hybrid approaches**

Each of the methodologies above addresses some but not all of the analytical features the Bitcoin case presents. DAPP has been integrated with economic evaluation across substantial flood-risk, coastal protection, and water management applications [57, 59, 60], with the standard integration using cost-benefit analysis or net present value scorecards across pathways. The combined approach I suggest as the most productive empirical follow-up extends this established integration with three structural components specific to the Bitcoin case.

First, the process would involve parallel DAPP analyses on the threat and response sides, with the threat-side treating quantum capability emergence as a process to be characterized through its own goals, milestones, and tasks rather than as an exogenous scenario input. Second, MACC-style cost decomposition nested within the response-side pathway could integrate MACC apparatus that serves the cost-effectiveness ranking task across candidate Bitcoin interventions. Third, a Bayesian representation of the threat-side that captures the conditional dependencies between capability milestones would support dynamic updating as evidence accumulates.

The integration would run through milestone-conditioned decision points: threat-side milestones trigger Bayesian updates that propagate through the network and supply the response-side decision tree with evolving probability distributions over capability states, while response-side milestones flag whether the Bitcoin community is on track relative to whatever timeline the updated threat-side analysis indicates. MACC analysis applies only on the response side because cost-effectiveness comparisons are meaningful only where genuine alternatives exist competing for the community's coordination capacity; the threat side does not warrant MACC analysis because the Bitcoin community is not selecting among quantum-research investments.

The Bayesian extension warrants explicit specification because the threat-side problem has structure that pure pathway analysis does not exploit. Capability milestones are not analytically independent: NIST FIPS 206 finalization conditions which signature schemes face deprecation risk; demonstrated factorization at lower key lengths conditions probability estimates for factorization at higher key lengths; and qubit-count and gate-cost improvements interact non-additively to compress CRQC arrival distributions. A Bayesian Belief Network (BBN), or an influence diagram extending the BBN with decision and utility nodes, represents these conditional dependencies directly and propagates evidence updates through the network when milestones are observed.

The Mosca/Piani threat timeline elicitations [26] are already Bayesian in spirit – expert judgments aggregated into probability distributions over CRQC arrival – and a network representation formalizes what the elicitation literature does informally. The substantive advantage is that the threat-side analysis becomes a continuously updatable inference engine rather than a static set of scenario inputs, which matches the temporal structure of a threat that will reveal itself incrementally over the relevant decision horizon.

Bayesian methods have been integrated with both DAPP and MACC frameworks in prior research. Jeong et al. [61] develop a risk-adaptive technology roadmap that converts an existing roadmap to a Bayesian network with adaptive plans, constructs conditional probability tables, and uses posterior probability inference to remap pathways as risk events occur, validating the approach on artificial intelligence as a technology domain. Buurman and Babovic [62] integrate adaptation pathways with real options analysis using Bayesian updating on observed events to drive sequential pathway decisions in climate adaptation contexts. Rodriguez-Sanchez et al. [63] develop a probabilistic MACC for smart-city mitigation strategies that integrates Monte Carlo simulation for parametric uncertainty with Bayesian network inference for conditional dependencies, producing risk metrics that quantify the probability of strategy failure under specified evidence configurations. These precedents establish the methodological tradition the Bitcoin application could extend; the architecture proposed here adds Bayesian-DAPP integration on the threat side, Bayesian-MACC integration on the response side, and the milestone-conditioned coupling between them.

5.5. Extensions beyond Bitcoin

The architecture developed here extends naturally beyond the Bitcoin case to any analytical problem where multiple technology pathways co-evolve and a focal domain must respond to milestones that arise exogenously. Quantum capability emergence is itself an instance of the pattern: it sits within a stack of co-dependent technological advances where progress at each level conditions what is feasible at the levels above, and where the same milestone reads as threat or benefit depending on the values framing the analytical question. Quantum is a cybersecurity threat to Bitcoin but a substantial opportunity across pharmaceutical research, materials science, and climate modeling.

Each component of the architecture generalizes, though their applicability varies by domain. Parallel DAPP analysis on threat and response sides applies wherever a focal governance domain must respond to capability emergence in an adjacent technological pathway. Bayesian representation of the threat side applies wherever capability milestones exhibit conditional dependencies that warrant dynamic updating as evidence accumulates, which characterizes most emerging-technology threats. MACC-style cost decomposition on the response side applies wherever genuine alternative interventions compete for the focal community's coordination capacity, which excludes domains where the response is dominated by a single dominant intervention. The deliberative lead-time formalization applies wherever the focal architecture's coordination latency materially conditions which pathways are reachable on the timescale the threat dictates – which characterizes most polycentric and federated governance domains and is most consequential where coordination is informal or norm-based rather than hierarchically supplied. The constituency-disaggregated MACC applies wherever the focal community contains analytically distinct welfare populations whose loss functions diverge over the response space, which characterizes any governance domain where uniform aggregation would bury rather than surface the values choice the analysis aims to inform.

6. Conclusion

Bitcoin's quantum dilemma is a coordination problem that exposes the directional structure of institutional latency. The same governance properties that make the architecture robust against capture, censorship, and arbitrary intervention – exit dominance, distributed authority, political commitments embedded in path-dependent informal norms – make the architecture slow to mobilize coordinated response when external conditions impose a deadline the architecture did not choose. BIP 360 progresses smoothly because it operates within the architecture's normal capacity for additive opt-in change. BIP 361 should be expected to encounter principled and persistent resistance because it forces engagement at the epistemic and political levels where Bitcoin's commitments meet the question of what those commitments are for.

The framework I have developed does not resolve the BIP 361 dispute, and the pragmatist question [14, 40] of what there is most reason to want – given conditions that did not exist when the original commitments were made – is one the Bitcoin community must answer through whatever deliberative process it can mobilize from informal norm-based resources. What the framework does is clarify the level at which the question must be answered and identify what the architecture does, and does not, supply for the deliberation that the answer requires. That is the institutional task the cryptography puts to Bitcoin, and recognizing it as such highlights the pressing need for deliberative tools to address the emerging, but deeply uncertain, quantum challenge

The analytical architecture I develop in this paper clarifies why Bitcoin's quantum migration is hard in the specific way that it is. Directional institutional latency and the four-level IAD analysis locate the BIP 360 / BIP 361 dispute at the epistemic level. That is where Bitcoin is structurally most resistant to coordinated change. The community must address the question the epistemic level forces before the political and implementation mechanisms can effectively respond: what are Bitcoin's commitments for, and is their service to that purpose negotiable under conditions the original commitments did not anticipate? The candidate methodologies identified here specify how subsequent empirical analysis could inform the deliberation the architecture forces but does not itself supply. Bitcoin's quantum-readiness depends on whether the community can mobilize deliberation and reach agreement on defensive measures, on the timescale the threat dictates, despite Bitcoin's lack of the hierarchical toolkit that other sectors have at their disposal.

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